

Part 2:

This is the MATLAB program code for the multi-band digital equalizer for speech clarity enhancement for audio files.

```
function [signal, fs] = loadAudio(fileName)
    % Input:
    %   fileName: String of file name within code directory
    %
    % Output:
    %   signal: Mono audio data normalised between -1 and 1
    %   fs: Sampling frequency

    try
        [raw, fs] = audioread(fileName);

        if size(raw, 2) > 1
            fprintf('Stereo signal detected. Converting to mono...\n');
            signal = mean(raw, 2);
        else
            signal = raw;
        end

        % Normalisation
        signal = signal / max(abs(signal));

        fprintf('Successfully loaded: %s\n', fileName);
        fprintf('Sampling Rate: %d Hz | Duration: %.2f seconds\n', fs, length(signal)/fs);
    catch err
        error('Error loading audio file: %s.', err.message);
    end
end

function bands = chooseMode()
    % Output:
    %   bands: [f0, f1, f2, ... fn]

    choice = 0;

    while choice < 1 || choice > 2
        fprintf('\n--- Equalizer Mode Selection ---\n');
        fprintf('1. Preset Mode\n');
```

```

        fprintf('2. Custom Mode\n');
        choice = input('Select mode (1 or 2): ');
    end

    if choice == 1
        % Preset Mode
        bands = [0, 100, 300, 800, 2000, 5000, 10000, 20000];
    else
        % Custome Mode
        n = input('Enter number of bands (between 5 and 10): ');
        while n < 5 || n > 10
            n = input('Invalid. Please enter a number between 5 and 10: ');
        end

        bands = zeros(1, n);
        bands(1) = 0;
        bands(end) = 20000;

        for i = 2 : n - 1
            msg = sprintf('Enter end frequency for Band %d in Hz (Must be > %d): ', i,
- 1, bands(i - 1));
            bands(i) = input(msg);

            % Validation: Ensure frequencies are increasing
            while bands(i) <= bands(i - 1)
                bands(i) = input('Error: Frequency must be higher than previous band.
Try again: ');
            end
        end
    end
end

function gains = inputGains(bands)
    % Prompts user for dB gain for each frequency band
    % Input:
    %   bands [output of chooseMode()]
    % Output:
    %   gains: [g0, g1, g2, ... gn]

    numBands = length(bands) - 1;
    gains = zeros(1, numBands);

    fprintf('\n--- Gain Configuration (in dB) ---\n');
    for i = 1:numBands
        prompt = sprintf('Band %d (%g Hz - %g Hz): ', i, bands(i), bands(i+1));
        gains(i) = input(prompt);
    end
end
end

```